Conditional Statements in Bash

Definition:

Conditional statements in Bash allow you to make decisions in scripts, executing specific commands based on whether a condition is true or false.

Types of Conditional Statements in Bash

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1. if Statement

Executes commands only if a condition is true.

Syntax:

```
if [ condition ]
then
    commands
fi
```

Example:

```
a=10
if [ $a -gt 5 ]; then
  echo "a is greater than 5"
fi
```

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2. if-else Statement

Executes one block if true, another if false.

Syntax:

```
if [ condition ]
```

then
  commands_if_true
else
  commands_if_false
fi

Example:
```
num=8
if [ $num -gt 10 ]; then
 echo "Number greater than 10"
else
 echo "Number less than or equal to 10"
fi
```

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3. if-elif-else Ladder

Used for multiple conditions.

Syntax:
```
if [ condition1 ]
then
  commands1
elif [ condition2 ]
then
  commands2
else
  commands3
fi
```

Example:
```
marks=75
if [ $marks -ge 90 ]; then
 echo "Grade: A"
```

```
elif [ $marks -ge 75 ]; then
  echo "Grade: B"
else
  echo "Grade: C"
fi
```

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4. Nested if Statements

Used when one if is inside another.

Syntax:

```
if [ condition1 ]
then
    if [ condition2 ]
    then
        commands
    fi
fi
```

Example:

```
a=10
b=20
if [ $a -lt 15 ]; then
  if [ $b -gt 15 ]; then
    echo "a < 15 and b > 15"
  fi
fi
```

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5. case Statement

Checks a variable against multiple patterns.

Syntax:

```
case $variable in
  pattern1)
      commands
      ;;
  pattern2)
      commands
      ;;
  *)
      default_commands
      ;;
esac
```

Example:

```
echo "Enter a number (1-3):"
read num
case $num in
  1) echo "You chose One" ;;
  2) echo "You chose Two" ;;
  3) echo "You chose Three" ;;
  *) echo "Invalid choice" ;;
esac
```

Common Operators:

Arithmetic: -eq, -lt, -gt, -ne

String: =, !=, -z, -n

File Test: -f, -d, -r, -w, -x

Example Combining All:

```
read -p "Enter your age: " age
if [ $age -lt 13 ]; then
  echo "Child"
```

```
elif [ $age -lt 20 ]; then
  echo "Teenager"
elif [ $age -lt 60 ]; then
  echo "Adult"
else
  echo "Senior"
fi
```

Summary Table

Type	Description	Example
if	Runs commands if true	if [$a -gt 10]; then ... fi
if-else	Executes one block if false	if [$x -eq 1]; else ... fi
if-elif-else	Multiple conditions	if ... elif ... else ... fi
Nested if	if inside another if	if [...]; if [...]; fi; fi
case	Pattern-based switch	case $var in ... esac